

Practical 11 — POSIX Threads and Mutex

AIM

To develop a multithreaded counter application using POSIX threads (`pthread_create()` and `pthread_join()`), demonstrate a race condition by allowing multiple threads to update a shared variable concurrently, and eliminate the race condition using mutex locks. The outputs are compared to analyze the effect of synchronization.

OBJECTIVES

1. Create multiple threads using `pthread_create()`.
2. Use `pthread_join()` to wait for thread completion.
3. Demonstrate a **race condition** on a shared counter.
4. Use `pthread_mutex_lock()` and `pthread_mutex_unlock()` to protect the shared variable.
5. Compare synchronized and unsynchronized outputs.
6. Analyze the impact of synchronization on the final counter value.

REQUIREMENTS

- Ubuntu/Linux
- GCC compiler
- POSIX Threads (`pthread`)

THEORY

POSIX Threads (Pthreads):

POSIX threads allow multiple threads to execute concurrently within the same process.

`pthread_create()`

Creates a new thread and starts its execution.

`pthread_join()`

Makes the main thread wait until the specified thread finishes.

Race Condition:

A race condition occurs when multiple threads access and modify the same shared variable at

the same time. Because the operations can overlap, the final result may be incorrect or unpredictable.

Mutex:

A mutex (mutual exclusion lock) allows only one thread at a time to access the critical section.

Functions used:

`pthread_mutex_lock()`

`pthread_mutex_unlock()`

ALGORITHM — WITHOUT MUTEX

1. Start the program.
2. Declare a shared counter variable.
3. Create multiple threads using `pthread_create()`.
4. Each thread increments the shared counter repeatedly.
5. Allow all threads to access the counter concurrently.
6. Wait for all threads using `pthread_join()`.
7. Display the final counter value.
8. Observe the race condition.
9. Stop.

ALGORITHM — WITH MUTEX

1. Start the program.
2. Declare a shared counter variable.
3. Initialize a mutex.
4. Create multiple threads.
5. Each thread locks the mutex before modifying the counter.
6. Increment the shared counter.
7. Unlock the mutex after the update.
8. Wait for all threads using `pthread_join()`.
9. Display the final counter value.
10. Destroy the mutex and stop.

EXPECTED COMPARISON

Version	Synchronizatio n	Result
Without mutex	 No	May produce an incorrect/unpredictable counter due to race condition

With mutex  Yes

Correct and consistent counter value

RESULT

Thus, a multithreaded counter application was successfully implemented using POSIX threads. The race condition caused by concurrent access to a shared variable was demonstrated, and synchronization using mutex locks was implemented successfully. The mutex-protected version provides controlled access to the shared counter and produces the expected result.

Final comparison for your record

Version	Mutex	Final Counter
Race condition	No	Varies
Synchronized	Yes	4,000,000

RESULT

Thus, a multithreaded counter application was successfully implemented using POSIX threads. The race condition caused by concurrent access to a shared variable was demonstrated. Mutex synchronization using `pthread_mutex_lock()` and `pthread_mutex_unlock()` was then implemented, producing the correct final counter value of 4,000,000.



kruthi@LAPTOP-LP9QU3RR: ~



```
/home/kruthi/p10
kruthi@LAPTOP-LP9QU3RR:~/p10$ ls
kruthi@LAPTOP-LP9QU3RR:~/p10$ echo "This is inode practical." > original.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ ls
original.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ ls -l original.txt
-rw-r--r-- 2 kruthi kruthi 25 Sep 28 08:35 original.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ ln original.txt hardlink.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ ls -l original.txt hardlink.txt
-rw-r--r-- 2 kruthi kruthi 25 Sep 28 08:35 original.txt
-rw-r--r-- 2 kruthi kruthi 25 Sep 28 08:35 hardlink.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ ln -s original.txt symlink.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ ls -l original.txt hardlink.txt symlink.txt
-rw-r--r-- 2 kruthi kruthi 25 Sep 28 08:35 original.txt
-rw-r--r-- 2 kruthi kruthi 25 Sep 28 08:35 hardlink.txt
lrwxrwxrwx 1 kruthi kruthi 12 Sep 28 08:36 symlink.txt -> original.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ find . -inum 2218
./hardlink.txt
./original.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ find . -inum 2219
./symlink.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ cat hardlink.txt
This is inode practical.
kruthi@LAPTOP-LP9QU3RR:~/p10$ cat symlink.txt
This is inode practical.
kruthi@LAPTOP-LP9QU3RR:~/p10$ echo "This file is used to demonstrate memory mapped I/O using mmap()." > source.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ cat source.txt
This file is used to demonstrate memory mapped I/O using mmap().
kruthi@LAPTOP-LP9QU3RR:~/p10$ nano mmap_copy.c
kruthi@LAPTOP-LP9QU3RR:~/p10$ gcc mmap_copy.c -o mmap_copy
kruthi@LAPTOP-LP9QU3RR:~/p10$ ls
hardlink.txt mmap_copy mmap_copy.c original.txt source.txt symlink.txt
kruthi@LAPTOP-LP9QU3RR:~/p10$ ./mmap_copy
File copied successfully using mmap().
kruthi@LAPTOP-LP9QU3RR:~/p10$ cat mmap_output.txt
This file is used to demonstrate memory mapped I/O using mmap().
kruthi@LAPTOP-LP9QU3RR:~/p10$ nano traditional_copy.c
kruthi@LAPTOP-LP9QU3RR:~/p10$ gcc traditional_copy.c -o traditional_copy
kruthi@LAPTOP-LP9QU3RR:~/p10$ ./traditional_copy
File copied successfully using read()/write().
kruthi@LAPTOP-LP9QU3RR:~/p10$ cat traditional_output.txt
This file is used to demonstrate memory mapped I/O using mmap().
kruthi@LAPTOP-LP9QU3RR:~/p10$ time ./mmap_copy
File copied successfully using mmap().

real    0m0.003s
user    0m0.002s
sys     0m0.001s
kruthi@LAPTOP-LP9QU3RR:~/p10$ time ./traditional_copy
File copied successfully using read()/write().

real    0m0.003s
user    0m0.000s
sys     0m0.002s
kruthi@LAPTOP-LP9QU3RR:~/p10$ cd ..
kruthi@LAPTOP-LP9QU3RR:~$ mkdir p11
kruthi@LAPTOP-LP9QU3RR:~$ pwd
/home/kruthi
kruthi@LAPTOP-LP9QU3RR:~$ ls
2520030340_kruthi  memory_allocation  program.c
ErrorLogger.class  memory_allocation.c  shell
InputValidator.class  os-operating-system-and-system-programming-OSSP  shell.c
OSSP                 os_practical5       shell.c.save
Prime-ossop-os       os_practical7       shell.c.save.1
Skill21.class        p10                 skill22
Skill21.java         p11                 skill24
UserService.class    p9                  student.txt
copy_on_write        pipeline_ipc         text.txt
copy_on_write.c      process.c.save
directory_name       program
kruthi@LAPTOP-LP9QU3RR:~$ nano race.c
kruthi@LAPTOP-LP9QU3RR:~$ gcc race.c -o race -pthread
kruthi@LAPTOP-LP9QU3RR:~$ ls
2520030340_kruthi  memory_allocation  program.c
ErrorLogger.class  memory_allocation.c  race
InputValidator.class  os-operating-system-and-system-programming-OSSP  race.c
OSSP                 os_practical5       shell
Prime-ossop-os       os_practical7       shell.c
Skill21.class        p10                 shell.c.save
Skill21.java         p11                 shell.c.save.1
UserService.class    p9                  skill22
copy_on_write        pipeline_ipc         skill24
copy_on_write.c      process.c.save      student.txt
directory_name       program             text.txt
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kruthi@LAPTOP-LP9QU3RR:~$ ls
2520030340_kruthi  memory_allocation  program.c
ErrorLogger.class  memory_allocation.c  race
InputValidator.class  os-operating-system-and-system-programming-OSSP  race.c
OSSP                 os_practical5       shell
Prime-ossop-os       os_practical7       shell.c
Skill21.class        p10                 shell.c.save
Skill21.java         p11                 shell.c.save.1
UserService.class    p9                  skill22
copy_on_write        pipeline_ipc         skill24
copy_on_write.c      process.c.save      student.txt
directory_name       program             text.txt
```



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InputValidator.class   os-operating-system-and-system-programming-OSSP  race.c
OSSP                   os_practical5          shell
Prime-oss-p-os         os_practical7          shell.c
Skill21.class          p10                    shell.c.save
Skill21.java           p11                    shell.c.save.1
UserService.class      p9                      skill22
copy_on_write          pipeline_ipc            skill24
copy_on_write.c        process.c.save          student.txt
directory_name         program                 text.txt
kruthi@LAPTOP-LP9QU3RR:~$ gcc race.c -o race -pthread
kruthi@LAPTOP-LP9QU3RR:~$ ls
2520030340_kruthi      memory_allocation      program.c
ErrorLogger.class      memory_allocation.c    race
InputValidator.class   os-operating-system-and-system-programming-OSSP  race.c
OSSP                   os_practical5          shell
Prime-oss-p-os         os_practical7          shell.c
Skill21.class          p10                    shell.c.save
Skill21.java           p11                    shell.c.save.1
UserService.class      p9                      skill22
copy_on_write          pipeline_ipc            skill24
copy_on_write.c        process.c.save          student.txt
directory_name         program                 text.txt
kruthi@LAPTOP-LP9QU3RR:~$ gcc race.c -o race -pthread
kruthi@LAPTOP-LP9QU3RR:~$ ./race
Final counter value: 2027328
kruthi@LAPTOP-LP9QU3RR:~$ ./race
Final counter value: 1445772
kruthi@LAPTOP-LP9QU3RR:~$ ./race
Final counter value: 1687170
kruthi@LAPTOP-LP9QU3RR:~$ ./race
Final counter value: 1655017
kruthi@LAPTOP-LP9QU3RR:~$ nano mutex.c
kruthi@LAPTOP-LP9QU3RR:~$ gcc mutex.c -o mutex -pthread
kruthi@LAPTOP-LP9QU3RR:~$ ./mutex
Final counter value: 4000000
kruthi@LAPTOP-LP9QU3RR:~$ |
```



kruthi@LAPTOP-LP9QU3RR: ~



```
kruthi@LAPTOP-LP9QU3RR:~$ ls
2520030340_kruthi      memory_allocation      program.c
ErrorLogger.class      memory_allocation.c    race
InputValidator.class   os-operating-system-and-system-programming-OSSP  race.c
OSSP                   os_practical5          shell
Prime-oss-p-os         os_practical7          shell.c
Skill21.class          p10                    shell.c.save
Skill21.java           p11                    shell.c.save.1
UserService.class      p9                      skill22
copy_on_write          pipeline_ipc            skill24
copy_on_write.c        process.c.save          student.txt
directory_name         program                 text.txt
kruthi@LAPTOP-LP9QU3RR:~$ gcc race.c -o race -pthread
kruthi@LAPTOP-LP9QU3RR:~$ ls
2520030340_kruthi      memory_allocation      program.c
ErrorLogger.class      memory_allocation.c    race
InputValidator.class   os-operating-system-and-system-programming-OSSP  race.c
OSSP                   os_practical5          shell
Prime-oss-p-os         os_practical7          shell.c
Skill21.class          p10                    shell.c.save
Skill21.java           p11                    shell.c.save.1
UserService.class      p9                      skill22
copy_on_write          pipeline_ipc            skill24
copy_on_write.c        process.c.save          student.txt
directory_name         program                 text.txt
kruthi@LAPTOP-LP9QU3RR:~$ gcc race.c -o race -pthread
kruthi@LAPTOP-LP9QU3RR:~$ ./race
Final counter value: 2027328
kruthi@LAPTOP-LP9QU3RR:~$ ./race
Final counter value: 1445772
kruthi@LAPTOP-LP9QU3RR:~$ ./race
Final counter value: 1687170
kruthi@LAPTOP-LP9QU3RR:~$ ./race
Final counter value: 1655017
kruthi@LAPTOP-LP9QU3RR:~$ nano mutex.c
kruthi@LAPTOP-LP9QU3RR:~$ gcc mutex.c -o mutex -pthread
kruthi@LAPTOP-LP9QU3RR:~$ ./mutex
Final counter value: 4000000
kruthi@LAPTOP-LP9QU3RR:~$ |
```